

Introduction to Data Science

Group assignment 2

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1 Exercise 1

We want to find the loss function by minimizing the negative log-likelihood (NLL). Let the dataset be $\{(x_i, y_i)\}_{i=1}^n$.

The conditional density is:

$$f_{Y|X}(y_i|x_i) = \frac{\lambda(x_i)^{y_i} e^{-\lambda(x_i)}}{y_i!}$$

where $\lambda(x_i) = \exp(\alpha \cdot x_i + \beta)$.

The loss $L(\alpha, \beta)$ is the negative log-likelihood:

$$\begin{aligned} L(\alpha, \beta) &= - \sum_{i=1}^n \ln(f_{Y|X}(y_i|x_i)) \\ &= - \sum_{i=1}^n \ln\left(\frac{\lambda(x_i)^{y_i} e^{-\lambda(x_i)}}{y_i!}\right) \\ &= - \sum_{i=1}^n [\ln(\lambda(x_i)^{y_i} e^{-\lambda(x_i)}) - \ln(y_i!)] \\ &= - \sum_{i=1}^n [\ln(\lambda(x_i)^{y_i}) + \ln(e^{-\lambda(x_i)})] + \sum_{i=1}^n \ln(y_i!) \\ &= - \sum_{i=1}^n [y_i \ln(\lambda(x_i)) - \lambda(x_i)] + C \\ &\quad (\text{where } C = \sum \ln(y_i!) \text{ is a constant independent of } \alpha \text{ and } \beta) \end{aligned}$$

Since we are minimizing with respect to α and β , we can drop the constant term C . This answers the note in the problem: we do not need the factorial term for the minimization.

With that, the loss function we need to minimize is simply:

$$\begin{aligned} L(\alpha, \beta) &= - \sum_{i=1}^n [y_i \ln(\lambda(x_i)) - \lambda(x_i)] \\ &= - \sum_{i=1}^n [y_i \ln(\exp(\alpha \cdot x_i + \beta)) - \exp(\alpha \cdot x_i + \beta)] \\ &= - \sum_{i=1}^n [y_i(\alpha \cdot x_i + \beta) - \exp(\alpha \cdot x_i + \beta)] \\ &= \sum_{i=1}^n (\exp(\alpha \cdot x_i + \beta) - y_i(\alpha \cdot x_i + \beta)) \end{aligned}$$

Thus, the loss to be minimized is:

$$L(\alpha, \beta) = \sum_{i=1}^n (\exp(\alpha \cdot x_i + \beta) - y_i(\alpha \cdot x_i + \beta))$$

2 Exercise 2

Let $X_1, \dots, X_n \sim \text{IID Uniform}(0, \theta)$. Let $\hat{\theta} = \max(X_1, \dots, X_n)$.

1. Distribution Function (CDF) of $\hat{\theta}$

Let y be in the support $[0, \theta]$. The CDF of $\hat{\theta}$ is $F_{\hat{\theta}}(y)$:

$$\begin{aligned} F_{\hat{\theta}}(y) &= P(\hat{\theta} \leq y) \\ &= P(\max(X_1, \dots, X_n) \leq y) \\ &= P(X_1 \leq y, X_2 \leq y, \dots, X_n \leq y) \end{aligned}$$

Since X_i are IID:

$$= \prod_{i=1}^n P(X_i \leq y)$$

The CDF for a single $X_i \sim \text{Uniform}(0, \theta)$ is $F_{X_i}(y) = \frac{y}{\theta}$

$$= \left(\frac{y}{\theta}\right)^n$$

So, $F_{\hat{\theta}}(y) = (y/\theta)^n$ for $0 \leq y \leq \theta$.

Probability Density Function (PDF) of $\hat{\theta}$

We differentiate the CDF to find the PDF, $f_{\hat{\theta}}(y)$:

$$f_{\hat{\theta}}(y) = \frac{d}{dy} F_{\hat{\theta}}(y) = \frac{d}{dy} \left(\frac{y}{\theta}\right)^n = n \left(\frac{y}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} = \frac{ny^{n-1}}{\theta^n}$$

2. Bias($\hat{\theta}$)

First, we compute the expected value of $\hat{\theta}$:

$$\begin{aligned} E[\hat{\theta}] &= \int_0^\theta y \cdot f_{\hat{\theta}}(y) dy \\ &= \int_0^\theta y \cdot \frac{ny^{n-1}}{\theta^n} dy \\ &= \frac{n}{\theta^n} \int_0^\theta y^n dy \\ &= \frac{n}{\theta^n} \left[\frac{y^{n+1}}{n+1} \right]_0^\theta \\ &= \frac{n}{\theta^n} \left(\frac{\theta^{n+1}}{n+1} \right) = \frac{n}{n+1} \theta \end{aligned}$$

The bias is defined as $\text{bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$:

$$\text{bias}(\hat{\theta}) = \frac{n}{n+1} \theta - \theta = \theta \left(\frac{n}{n+1} - 1 \right) = \theta \left(\frac{n - (n+1)}{n+1} \right) = -\frac{\theta}{n+1}$$

3. Standard Error $\text{se}(\hat{\theta})$

The standard error is $\text{se}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$. We first find $\text{Var}(\hat{\theta}) = E[\hat{\theta}^2] - (E[\hat{\theta}])^2$.

We need $E[\hat{\theta}^2]$:

$$\begin{aligned}
 E[\hat{\theta}^2] &= \int_0^\theta y^2 \cdot f_{\hat{\theta}}(y) dy \\
 &= \int_0^\theta y^2 \cdot \frac{ny^{n-1}}{\theta^n} dy \\
 &= \frac{n}{\theta^n} \int_0^\theta y^{n+1} dy \\
 &= \frac{n}{\theta^n} \left[\frac{y^{n+2}}{n+2} \right]_0^\theta \\
 &= \frac{n}{\theta^n} \left(\frac{\theta^{n+2}}{n+2} \right) = \frac{n}{n+2} \theta^2
 \end{aligned}$$

Now, compute the variance:

$$\begin{aligned}
 \text{Var}(\hat{\theta}) &= E[\hat{\theta}^2] - (E[\hat{\theta}])^2 \\
 &= \frac{n}{n+2} \theta^2 - \left(\frac{n}{n+1} \theta \right)^2 \\
 &= \theta^2 \left(\frac{n}{n+2} - \frac{n^2}{(n+1)^2} \right) \\
 &= \theta^2 \left(\frac{n(n+1)^2 - n^2(n+2)}{(n+2)(n+1)^2} \right) \\
 &= \theta^2 \left(\frac{n(n^2 + 2n + 1) - n^3 - 2n^2}{(n+2)(n+1)^2} \right) \\
 &= \theta^2 \left(\frac{n^3 + 2n^2 + n - n^3 - 2n^2}{(n+2)(n+1)^2} \right) \\
 &= \frac{n\theta^2}{(n+2)(n+1)^2}
 \end{aligned}$$

Finally, the standard error is:

$$\text{se}(\hat{\theta}) = \sqrt{\frac{n\theta^2}{(n+2)(n+1)^2}} = \frac{\theta\sqrt{n}}{(n+1)\sqrt{n+2}}$$

4. $\text{MSE}_n(\hat{\theta})$

The Mean Squared Error is $\text{MSE}_n(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{bias}(\hat{\theta}))^2$:

$$\begin{aligned}\text{MSE}_n(\hat{\theta}) &= \frac{n\theta^2}{(n+2)(n+1)^2} + \left(-\frac{\theta}{n+1}\right)^2 \\ &= \frac{n\theta^2}{(n+2)(n+1)^2} + \frac{\theta^2}{(n+1)^2} \\ &= \theta^2 \left(\frac{n}{(n+2)(n+1)^2} + \frac{1}{(n+1)^2} \right)\end{aligned}$$

Find a common denominator:

$$\begin{aligned}&= \theta^2 \left(\frac{n + (n+2)}{(n+2)(n+1)^2} \right) \\ &= \theta^2 \left(\frac{2n+2}{(n+2)(n+1)^2} \right) \\ &= \theta^2 \left(\frac{2(n+1)}{(n+2)(n+1)^2} \right) \\ &= \frac{2\theta^2}{(n+2)(n+1)}\end{aligned}$$

3 Exercise 3

We are given the continuous distribution with density:

$$p(x) = \frac{1}{2} \cos(x), \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

(a) Find the distribution function F

The distribution function $F(x)$ is the integral of the density $p(t)$ from the lower bound of the support $(-\pi/2)$ up to x .

$$\begin{aligned}F(x) &= \int_{-\pi/2}^x p(t) dt \\ &= \int_{-\pi/2}^x \frac{1}{2} \cos(t) dt \\ &= \frac{1}{2} [\sin(t)]_{-\pi/2}^x \\ &= \frac{1}{2} \left(\sin(x) - \sin\left(-\frac{\pi}{2}\right) \right) \\ &= \frac{1}{2} (\sin(x) - (-1)) \\ &= \frac{1}{2} (\sin(x) + 1)\end{aligned}$$

The full CDF is:

$$F(x) = \begin{cases} 0 & x \leq -\pi/2 \\ \frac{1}{2}(\sin(x) + 1) & -\pi/2 < x < \pi/2 \\ 1 & x \geq \pi/2 \end{cases}$$

(b) Find the inverse distribution function F^{-1}

To find the inverse F^{-1} , we set $y = F(x)$ for $x \in (-\pi/2, \pi/2)$ and solve for x . The range of $F(x)$ for this domain is $y \in (0, 1)$.

$$\begin{aligned} y &= F(x) \\ y &= \frac{1}{2}(\sin(x) + 1) \\ 2y &= \sin(x) + 1 \\ 2y - 1 &= \sin(x) \\ x &= \arcsin(2y - 1) \end{aligned}$$

The range of the arcsin function is $[-\pi/2, \pi/2]$, which matches the domain of x we are inverting. By convention, we express the inverse function in terms of x .

$$F^{-1}(x) = \arcsin(2x - 1), \quad \text{for } x \in (0, 1)$$

(c) Find a density g and a value M

For an Accept-Reject sampler, we need a proposal density $g(x)$ and a constant $M > 0$ such that $p(x) \leq Mg(x)$ for all x .

Let's choose $g(x)$ to be a $\text{Uniform}(-\pi/2, \pi/2)$ density. This is a simple choice because its support matches the support of $p(x)$. The density for $g(x)$ is:

$$g(x) = \frac{1}{b-a} = \frac{1}{\pi/2 - (-\pi/2)} = \frac{1}{\pi}$$

Now we must find the smallest M that satisfies the condition $p(x) \leq Mg(x)$:

$$\begin{aligned} \frac{1}{2} \cos(x) &\leq M \cdot \frac{1}{\pi} \\ \text{Multiply by } \pi \text{ to isolate } M : \\ \frac{\pi}{2} \cos(x) &\leq M \end{aligned}$$

This inequality must hold for all $x \in (-\pi/2, \pi/2)$. To find the smallest M , we must find the maximum value of the left-hand side:

$$M \geq \max_{x \in (-\pi/2, \pi/2)} \left(\frac{\pi}{2} \cos(x) \right)$$

The function $\cos(x)$ achieves its maximum value of 1 at $x = 0$.

$$\max \left(\frac{\pi}{2} \cos(x) \right) = \frac{\pi}{2} \cdot \cos(0) = \frac{\pi}{2}$$

Therefore, the smallest possible value for M is $\frac{\pi}{2}$.

We have:

$$g(x) = \frac{1}{\pi}, \quad M = \frac{\pi}{2}$$

4 Exercise 4

In the first part of the exercise we have to show that X_n is in fact a Markov chain, so that it satisfies the Markov property. This property formulated for discrete variables can be written in the following way.

Markov property.

$$P(X_{t+1} = x_{t+1} \mid X_t = x_t, X_{t-1} = x_{t-1}, \dots, X_0 = x_0) = P(X_{t+1} = x_{t+1} \mid X_t = x_t)$$

which pretty much means that the future of the process is only determined by its present and not the past.

When we look at X_n , we can observe that it is a non decreasing sequence, so that the next value can only be equal or higher than the last value of the sequence. In other words, we can write that

$$X_{n+1} = \max((Y_1, Y_2, \dots, Y_n), Y_{n+1})$$

so

$$X_{n+1} = \max(X_n, Y_{n+1})$$

When written this way, we can see that the next value in our sequence depends solely on the current state of the system. Moreover, Y_{n+1} also does not depend on the previous elements in the sequence, as they are independent of each other, therefore, X_n satisfies the Markov property, so it is a Markov chain.

Transition matrix

In the next part of this exercise, we have to find a transition matrix P of X_n that represents all the probabilities of moving between states. We have 4 possible states in our sequence, so P will be a 4x4 matrix. With $p_0 = 0.1$, $p_1 = 0.3$, $p_2 = 0.2$, $p_3 = 0.4$ and ordering states as 0, 1, 2, 3, we can fill out the matrix by looking at different scenarios eg. when in state 0, the probability of moving to 0 again is exactly 0.1, as $P(Y_n = 0) = 0.1$, so $P_{00} = 0.1$. Similarly, moving to state 1 has probability 0.3, so $P_{01} = 0.3$ etc. However, when $X_n = 1$, $P(X_{n+1} = 0 \mid X_n = 1) = 0$, as X_n is a non decreasing sequence. In that case, no matter if Y_{n+1} is 0 or 1, X_{n+1} will be 1, so $P(X_{n+1} = 1 \mid X_n = 1) = 0.1 + 0.3 = 0.4$. Following this logic, we can fill out the whole transition matrix P , which will look like this:

$$P = \begin{bmatrix} 0.1 & 0.3 & 0.2 & 0.4 \\ 0 & 0.4 & 0.2 & 0.4 \\ 0 & 0 & 0.6 & 0.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5 Exercise 5

The problem asks for two things:

1. An estimate of the p -quantile q using the empirical distribution function (EDF) $\hat{F}_n$.
2. A confidence interval for q using the Dvoretzky-Kiefer-Wolfowitz (DKW) inequality.

1. Estimate of the p -Quantile (q)

The true p -quantile q is defined as the smallest value x such that the cumulative distribution function (CDF) $F(x)$ is at least p :

$$q = \inf\{x : F(x) \geq p\}$$

We are asked to find a "plug-in" estimator using the empirical distribution function (EDF), $\hat{F}_n$. The EDF is defined as:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x)$$

where $I(\cdot)$ is the indicator function.

We replace the true CDF F with the EDF $\hat{F}_n$ in the definition of the quantile to get the estimator $\hat{q}$:

$$\hat{q} = \inf\{x : \hat{F}_n(x) \geq p\}$$

Let $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ be the order statistics of the sample (the data sorted from smallest to largest). The EDF $\hat{F}_n(x)$ is a step function that jumps by $1/n$ at each $X_{(i)}$. The value $\hat{F}_n(X_{(k)})$ is k/n .

We are looking for the smallest x such that $\hat{F}_n(x) \geq p$. This occurs at the smallest order statistic $X_{(k)}$ for which $k/n \geq p$. This k is the smallest integer satisfying $k \geq np$, which is the definition of the ceiling function.

$$k = \lceil np \rceil$$

Therefore, the plug-in estimate for the p -quantile is the element at the k -th index of the sorted sequence of samples:

$$\hat{q} = X_{(k)} \quad \text{where} \quad k = \lceil np \rceil$$

2. Confidence interval for q via DKW

From the DKW inequality, we obtain an upper-bound on the probability that worst-case error in the EDF approximation of the CDF exceeds some ϵ , as given by:

$$P(\sup_x |\hat{F}_n(x) - F(x)| > \epsilon) \leq 2e^{-2n\epsilon^2}$$

Letting $\alpha = 2e^{-2n\epsilon^2}$, the DKW gives us a $1 - \alpha$ confidence interval (CI), such that with probability $1 - \alpha$:

$$\hat{F}_n(x) - \epsilon \leq F(x) \leq \hat{F}_n(x) + \epsilon$$

Thus, we find the empirical quantiles at levels $p \pm \epsilon$:

$$L_n = \hat{F}_n^{-1}(p - \epsilon) = \inf\{x : \hat{F}_n(x) \geq p - \epsilon\}$$

and

$$U_n = \hat{F}_n^{-1}(p + \epsilon) = \inf\{x : \hat{F}_n(x) \geq p + \epsilon\}$$

We note that L_n corresponds to the first order statistic $X_{(k_1)}$ where k_1 takes the smallest possible integer value between 1 and n , such that $k_1/n \geq p - \epsilon$.

Similarly, U_n corresponds to the first order statistic $X_{(k_2)}$ where k_2 takes the smallest possible integer value between 1 and n , such that $k_2/n \geq p + \epsilon$.

Thus, our $(1 - \alpha)$ confidence interval is given by:

$$[X_{(k_1)}, X_{(k_2)}]$$

where

$$k_1 = \max \{1, \lceil n(p - \epsilon) \rceil\}, \quad k_2 = \min \{n, \lceil n(p + \epsilon) \rceil\}$$

and

$$\epsilon = \sqrt{\frac{1}{2n} \ln \left(\frac{2}{\alpha} \right)}$$